

Data analytics numerical questions

data analytics (KIET Group of Institutions)

Set of Questions

Q.1 Find the mean, median, and mode for the following data:
85,75,30,86,85,30,35,45,85

Q.2 Find 5 number summary on the following data
34, 18, 100, 27, 54, 52, 93, 59, 61, 87, 68, 85, 78, 82, 91

Q.3 Find the outlier using whisker box plot
60,82,82,88,84,92,90,90,97,93

Q.4 Write the significance of Correlation. Calculate the coefficient of correlation between the height (in inches) of father and height of son (in inches) from the following data:
Height of father: 66 67 68 69 70 71 72
Height of son : 68 69 67 69 72 70 74

Q.5 Write the significance of regression. The following data relate to number of days and weight gain. Find regression of y on x
Time(days): 3 4 5 6 7 8 9 10
Weight(y) 1.4 1.5 2.2 2.4 3.1 3.2 3.2 3.9

Q.6 Find the intercept and slope in the linear regression based on the following data.

X	Y
1	5
3	3
4	2
5	0
8	0

Q.7 Given below are the monthly income and their net savings of a sample of 10 supervisory staff belonging to a firm. Calculate the correlation coefficient.

Monthly income(Rs.): 780 360 980 250 750 820 900 620 650 390

Net Savings: 84 51 91 60 68 62 86 58 53 47

$r = 0.78$ approx.

Q.8

Detect the outlier if any from the given dataset using a whisker box plot.

Month	Rainfall (in mm)
January	50
February	55
March	60
April	45
May	65
June	70
July	80
August	85

September	90
October	95
November	100
December	105

Q.9 The decrease in heart rate (beats/min) due to different concentrations of a drug are given below. Find the equation for the regression using the method of least square

Dose of drug(mg)	1.0	1.25	1.50	1.75	2.0	2.25	2.50	2.75	3.0
Decrease in heartbeat	10	12	12	14	16	17	20	18	21

Q.10

TID	Refund	Marital Status	Taxable Income	Cheat
1	Yes	Single	125 K	No
2	No	Married	100 K	No
3	No	Single	70 K	No
4	Yes	Married	120 K	No
5	No	Divorced	95 K	Yes
6	No	Married	60 K	No
7	Yes	Divorced	220 K	No
8	No	Single	85 K	Yes
9	No	Married	75 K	No
10	No	Single	90 K	Yes

Classify the following data using Naïve Bayes Algorithm

Refund	Marital Status	Taxable Income	Cheat
No	Married	80 K	?

Q.11 Given data= {3,4,5,6,7,9; 2,6,5,7,8,9}. Compute the principal component using PCA algorithm.

Q.12 For Data

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Classify the tuple(x')=(Outlook=*Sunny*, Temperature=*Cool*, Humidity=*High*, Wind=*Strong*)

Q.13 Compute the absolute error, mean square error and root mean square error based on the following data.

Observed values (actual): [10, 15, 20, 25, 30]

Predicted values: [12, 18, 22, 24, 28]

Q.14 Normalized the Data: [15, 25, 30, 40, 50] **applying** standard normalization

Q.15 Normalized the Data: [10, 20, 30, 40, 50] applying Min Max normalization

Q.16 Evaluate the performance of the classification model using the following confusion matrix.

	Cat	Dog	Rabbit
Cat	5	2	0
Dog	3	3	2
Rabbit	0	1	11

Q.17 For the following dataset

Transaction ID	Item Purchased
	Bread, Milk,Eggs
	Bread, Diapers,Beer,Eggs
	Milk,Diapers,Beer,Coke
	Bread, Milk,Diapers, Beer
	Bread, Milk, Diapers,Coke

- Apply the Apriori algorithm to find all frequent itemsets with a minimum support of 40%.
- Generate association rules with a minimum confidence of 60%.
- Interpret the results obtained from the association mining process and provide insights into the purchasing patterns of customers.

Q.18 Apply market basket analysis on given transaction database.

Transaction	Item
T ₁	Bread, Jelly , PeanutButter
T ₂	Bread, PeanutButter
T ₃	Bread, Milk, PeanutButter
T ₄	Jam, Bread
T ₅	Jam, Milk

Determine support and confidence for the following association rule

(i) PeanutButter $\rightarrow$ Bread

(ii) Bread $\rightarrow$ PeanutButter

Q.19 Estimate the different moments for the following data stream.

m, n, c, n, d, m, c, d, m, n, d, c, m, m, n

Q.20 Suppose we have a stream of integers, and we want to estimate the number of distinct integers in the stream using the Flajolet-Martin algorithm. Let's say that the stream contains the following 10 integers:

4, 7, 2, 9, 4, 6, 2, 8, 1, 3

A hash function is $h(x) = \text{value mod } 16$

Q.21 Explain DGIM algorithm (Datar-Gionis-Indyk-Motwani Algorithm) using suitable example.

Q.22 Write the formula to calculate Euclidean distance and Manhattan distance between two points (x_1, y_1) and (x_2, y_2)

Q.23 Use the k-means algorithm and Euclidean distance to cluster the following 8 examples into 3 clusters:

$P_1 = (3, 12)$, $P_2 = (3, 7)$, $P_3 = (9, 5)$, $P_4 = (6, 10)$, $P_5 = (8, 6)$, $P_6 = (7, 5)$, $P_7 = (2, 3)$, $P_8 = (5, 11)$. Suppose that the initial seeds (centres of each cluster) are P_3 , P_6 , and P_8 . Run the K-means algorithm for 1 epoch only.

Q.24 Apply an agglomerative hierarchical algorithm to identify the different clusters.

	City1	City 2	City 3	City 4	City 5
City1	0				
City2	0.14	0			
City3	0.16	0.17	0		
City4	0.39	0.17	0.17	0	
City5	0.36	0.16	0.29	0.31	0

Q.25 Find a proximity matrix then apply an agglomerative hierarchical algorithm to get different clusters.

Student_ID	Marks
1	10
2	7
3	28
4	20
5	35